

SOP-SG-03 — Tech & Integration Selection

Kode	SOP-SG-03
Pemilik	Sogeking (Solution Architect)
Revisi	1.0 · Berlaku 2026-05-29 · Reviu per kuartal
Referensi	TOGAF ADM (fase D/E), tools/trade-off-matrix.md, tools/build-buy-partner.md
COBIT	APO03 (Managed Enterprise Architecture)

1. Tujuan

Memastikan setiap pemilihan stack/vendor/pola integrasi diambil dari **≥2 opsi nyata** (anti vendor-hype, anti 1-opsi), lewat **filter assessment @senku** yang wajib, ber-trade-off matrix, dengan keputusan **build/buy/partner** eksplisit dan **vendor lock-in di-treat otomatis Type-1**.

2. Ruang Lingkup

- **Berlaku:** pilih/ganti stack, vendor/SaaS, pola integrasi lintas sistem, build-vs-buy-vs-partner.
- **Tidak berlaku:** pilihan library lokal reversible dalam stack default tim (cukup di code review @kakashi).

3. Definisi & Istilah

- **Trade-off matrix** — tabel banding opsi per kriteria (cost/scale/security/maintainability/lock-in).
- **Build / Buy / Partner** — bikin sendiri / beli vendor / kerja sama, basis trade-off.
- **Vendor lock-in** — ketergantungan yang mahal/mustahil di-undo → **otomatis Type-1**.
- **Type-1 / Type-2** — irreversible (pintu 1-arah) vs reversible.

4. Referensi

TOGAF ADM fase **D** (Technology Architecture), **E** (Opportunities & Solutions — build-vs-buy). Reversibility (Bezos one-way/two-way door). Tata rule: **gak boleh keputusan 1-opsi**. Filter wajib @senku (assessment tech baru). Stack default tim: React + Zustand + Chakra v2 + Fauxbase.

5. Tanggung Jawab (RACI)

Aktivitas	R	A	C	I
Define problem + constraint	Sogeking	Sogeking	@lelouch	tim
Generate ≥ 2 opsi + trade-off matrix	Sogeking	Sogeking	@kakashi	tim
Filter assessment tech	@senku	Sogeking	@senku	tim
Propose keputusan	Sogeking	Sogeking	@kakashi, @senku	tim
Keputusan Type-1 (lock-in dll)	—	@tata	Sogeking, @senku	tim

6. Prosedur

6.1 Framing (TOGAF D)

- 6.1.1 Definisikan **problem + constraint** (apa yang dicapai, batas biaya/skala/UX, mandate F-1/F-2).
- 6.1.2 Cek **reuse dulu** — apakah stack default (Fauxbase dkk) sudah cukup? Anti rebuild.

6.2 Eksplorasi opsi

- 6.2.1 Generate **≥ 2 opsi nyata** (gak boleh 1-opsi, anti vendor-hype). Exit kalau cuma ketemu 1 → cari minimal 1 alternatif (termasuk "do nothing" / build sendiri).
- 6.2.2 **MANDATORY filter @senku** — tiap opsi tech baru wajib lewat assessment Senku (maturity, security, fit). Exit kalau belum di-assess → stop, kirim ke @senku dulu (kontrol **SG-C3**).

6.3 Trade-off & keputusan (TOGAF E)

- 6.3.1 Tentukan jalur **build / buy / partner** per build-buy-partner: Build = core differentiator + reversible; Buy = commodity + vendor mature + ROI>build; Partner = complementary + both gain.
- 6.3.2 Susun **trade-off matrix** (trade-off-matrix) per kriteria: **cost · scale · security · maintainability · lock-in**. Contoh *wedding app*: payment gateway → buy (commodity) vs build (over-engineering, langgar YAGNI).
- 6.3.3 **Decision point — reversibility**: ada **vendor lock-in / data residency / ganti stack besar?** → **otomatis Type-1** . Wajib ADR + **escalate @tata** (via Pre-Tata Gate @kakashi) + exit strategy, tunggu sign-off. Reversible (Type-2 ) → propose & lanjut.
- 6.3.4 Tulis rationale + arsipkan trade-off matrix; broadcast keputusan + alasan ke tim.

7. Pengecualian

- **Trial / Assess (tech radar)**: boleh coba tech di lingkungan terbatas tanpa keputusan final, **asal** belum di-lock ke produksi & sudah lewat filter @senku. Promote ke "Adopt" wajib lewat SOP ini lengkap.

8. Rekaman & Retensi

Rekaman	Lokasi	Retensi
Trade-off matrix	arch/ / log.md	Per seleksi
Assessment @senku	log.md	Per seleksi
ADR (Type-1 / lock-in)	adr/NNN-*.md	Permanen
Sign-off Type-1 @tata	ADR (field Escalation)	Permanen

9. Indikator Kinerja (KPI)

KPI	Rumus	Target
Keputusan 1-opsi	# seleksi tanpa alternatif	0
Trade-off-backed selection	# seleksi ber-trade-off matrix ÷ total	100%
Filter @senku coverage	# opsi tech baru lewat assessment ÷ total	100%
Vendor lock-in ber-ADR + exit plan	# lock-in ber-ADR ÷ total lock-in	100%

10. Riwayat Revisi

Rev	Tanggal	Perubahan
1.0	2026-05-29	Terbit pertama